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MILITARY SPECIFICATION
FOR
PROGRAMMER, RANGE AND AIRFRAME SEPARATION
MARK 3 MOD 1

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of Programmer, Range and Airframe Separation Mark 3 Mod 1. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form apart of this specification to the extent specified herein, except as noted.

SPECIFICATIONS

MILITARY

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels

FSC-1340

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MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 497947	Container, Ignition and Separation Assembly, Mark 176 Mod 1 and all documents listed thereon
LD 541505	Tester, Acceptance, Range and Airframe Separation Programmer and all documents listed thereon
LD 541536	Programmer, Range and Airframe Separation Mark 3 Mod 1 and all documents listed thereon
LD 541554	Preserved (Packaging Condition) (Programmer, Range and Airframe Separation) and all documents listed thereon
LD 541555	Shipping (Packing Condition) (For one Programmer, Range and Airframe Separation) and all documents listed thereon

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 12995	Operation and Maintenance Instructions for Range and Airframe Separation Programmer Acceptance Tester

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the programmer shall be in accordance with the requirements specified herein and with the applicable documents listed in BuWeps LD 541536.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the programmer shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The programmer shall meet the following performance and product characteristics.

3.3.1 Circuit resistance. The programmer circuits, listed in Table I, shall have the resistances indicated.

Table I. Programmer Circuit Resistance Requirements

Circuits			Test Equip. Specified in 4.4.1	Resistance in Ohms
Connector 1A1J1	Connector 1A1J2	Header		
S and L (S negative)			a	100 ± 10
*S and N (S negative)			a	910 ± 100
S and M (S positive)			a	25 to 200
S and F (S negative)			a	3700 ± 800
S and B (S negative)			a	3700 ± 800
S and d (S negative)			a	500 min
**S and f (S negative)			a	50 to 5000
		1A1J3 1 and 2	b	0.020 to 0.100
		1A1J6 1 and 2	b	0.020 to 0.100
BB and AA			b	2.20 to 2.75
BB and BC			b	2.20 to 2.75
BB		1A1J5 3	b	0.05 max
BA		1A1J5 4	b	0.05 max
BC		1A1J5 5	b	0.05 max

* Programmer cover side down
** The resistance between S and f (S negative) shall be 25 to 5000 ohms after performing the acceleration tests in 4.5.1.7.

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Table I. Programmer Circuit Resistance Requirements (Continued)

Circuits		Header	Test Equip. Specified in 4.4.1	Resistance in Ohms
Connector 1A1J1	Connector 1A1J2			
BB	H		b	0.05 max
BA	J		b	0.05 max
BC	L		b	0.05 max
b	A		b	0.075 max
h	C		b	0.075 max
e	D		b	0.075 max
b and a			b	0.04 max
E	E		b	0.075 max
J	F		b	0.075 max
AC and AB			b	0.725 ± 0.125
K	N		b	0.075 max
AA and AB			b	0.725 ± 0.125
S and H			b	0.15 max
P	K		b	0.075 max
c	P		b	0.075 max
S and g			b	0.15 max
H and C			b	0.04 max
S to housing			b	0.15 max
V and W			b	0.04 max
f and G			b	0.04 max
V	B		b	0.075 max

3.3.2 Insulation resistance. At a potential of 500 plus or minus 50 volts DC, the programmer shall have an insulation resistance of not less than 20 megohms between:

a. Each of the eight indendent circuits of connector 1A1J2
b. The eight independent circuits of connector 1A1J2 and the programmer housing
c. The eight independent circuits of connector 1A1J2 and the shell of connector 1A1J2

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3.3.3 Nonfunction indication.

3.3.3.1 Arming circuit. The resistance between contacts S and D of connector 1A1J1 shall be not less than 80 ohms, prior to being subjected to the acceleration test of 4.5.1.7. After the acceleration test the resistance shall be not more than 80 ohms.

3.3.3.2 Interlock circuit. The resistance between contacts S and A of connector 1A1J1 shall be not less than 80 ohms.

3.3.3.3 Misfire indicator circuit. The insulation resistance between contacts Y and U of connector 1A1J1 shall be not less than 5 megohms, when the applied potential is 500 plus or minus 50 volts DC.

3.3.4 Operation. The programmer, when energized with 12.75 plus or minus 0.25 volts DC applied across contacts f (positive) and H (negative) of connector 1A1J1, shall provide the following voltages, measured with respect to contact S of connector 1A1J1.

3.3.4.1 Regulated voltage. The programmer shall provide a regulated voltage of 10.000 plus or minus 0.010 volts DC, measured at pin M (positive).

3.3.4.2 Resaturation voltage. The programmer shall provide a voltage of 0.210 plus or minus 0.010 volt DC, measured at pin L (positive).

3.3.4.3 Twenty-eight volt output. The programmer shall provide 28 plus or minus 3 volts DC, measured at pin d (positive).

3.3.4.4 Channel one output. The programmer shall provide a readback voltage of 4 volts DC, across a resistance of 27 ohms, measured at pin D (positive) when a proper sequence of presaturation voltage, set-in voltage, and resaturation voltage is applied at pin F. (The sequence and values of these voltages are described in detail in 4.6.1.4.4.) Readback time shall be as specified in Table II.

3.3.4.5 Channel two output. The programmer shall provide a readback voltage of 4 volts DC across a resistance of 27 ohms measured at pin A when a proper sequence of presaturation voltage, set-in voltage, and resaturation voltage is applied at pin B. (The sequence and values of these voltages are described in detail in 4.6.1.4.5.) Readback time shall be as specified in Table II.

3.3.4.6 Performance at specific temperatures. The programmer shall provide channel one and channel two outputs in accordance with requirements of 3.3.4.4, 3.3.4.5 and Table II after being stabilized for 4 hours minimum at the indicated temperatures.

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Table II. Performance at Specific Temperatures

Set-in voltage (negative volts DC)	Required readback time (seconds)	Tolerance of readings as a maximum RMS value of error of readback time (seconds) at following temperatures		
		35 plus 5 minus 0 degrees F	Room ambient	120 plus 0 minus 5 degrees F
Channel one:				
25.000 ± 0.025	1.000	0.04	0.02	0.04
50.000 ± 0.050	2.000	0.04	0.02	0.04
100.000 ± 0.100	4.000	0.04	0.02	0.04
Channel two:				
20.000 ± 0.020	10.000	0.50	0.50	0.50
40.000 ± 0.040	20.000	0.50	0.50	0.50
80.000 ± 0.080	40.000	0.50	0.50	0.50

3.3.4.7 Output energy. The programmer shall provide output signals from the channel one and channel two functions as electrical pulses of not less than 2 million ergs when dissipated within 3 milliseconds through a resistance of 2.0 ohms.

3.3.4.8 Channel one output guarantee. The programmer shall provide a channel one output even though the signal from the accelerometer may cease before channel one has functioned.

3.3.5 Enclosure. The programmer shall be airtight with a leakage rate of not more than 10 to the minus fifth power cubic centimeter per second of helium at one atmosphere pressure differential.

3.4 Environmental requirements. Unless otherwise specified, the programmer shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

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3.4.1 Temperature, storage condition. The programmer shall withstand temperatures from minus 65 to plus 160 degrees Fahrenheit (F).

3.4.2 Humidity. The programmer shall withstand relative humidity up to 95 percent as specified in MIL-E-5272, Procedure I.

3.4.3 Shock, standby condition. The programmer shall withstand shocks of 40g (g = unit of gravitational acceleration) peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 5 milliseconds.

3.4.4 Vibration, standby condition. The programmer shall withstand vibration as specified for Type I equipment in MIL-STD-167 up to 25 cps (cycles per second).

3.4.5 Vibration, operating condition. The programmer shall withstand vibration through a frequency range of 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less. During vibration the programmer shall meet the channel one and channel two output requirements of 3.3.4.4 and 3.3.4.5. The individual readback times shall be within 0.05 and 0.80 second of the required readback times for channel one and channel two, respectively.

3.4.6 Temperature, operating condition. The programmer shall meet the performance and product characteristics while at temperatures from 35 to 120 degrees F.

3.4.7 Acceleration. The programmer shall withstand an acceleration of 25g for 4 seconds applied in the direction shown in Figure 1, and thereafter shall meet the requirements of 3.3.1, 3.3.3.1 and 3.3.6.

3.4.8 Operating life. The programmer shall meet the operational requirement of 3.3.4 periodically during 500 hours of standby operation and thereafter shall meet the performance and product characteristics specified herein.

3.4.9 Fungus. The programmer shall withstand fungus conditions as specified in MIL-E-5272, Procedure I.

3.4.10 Salt spray. The programmer shall withstand salt spray conditions as specified in MIL-E-5272, Procedure I, for 50 hours.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduct ion sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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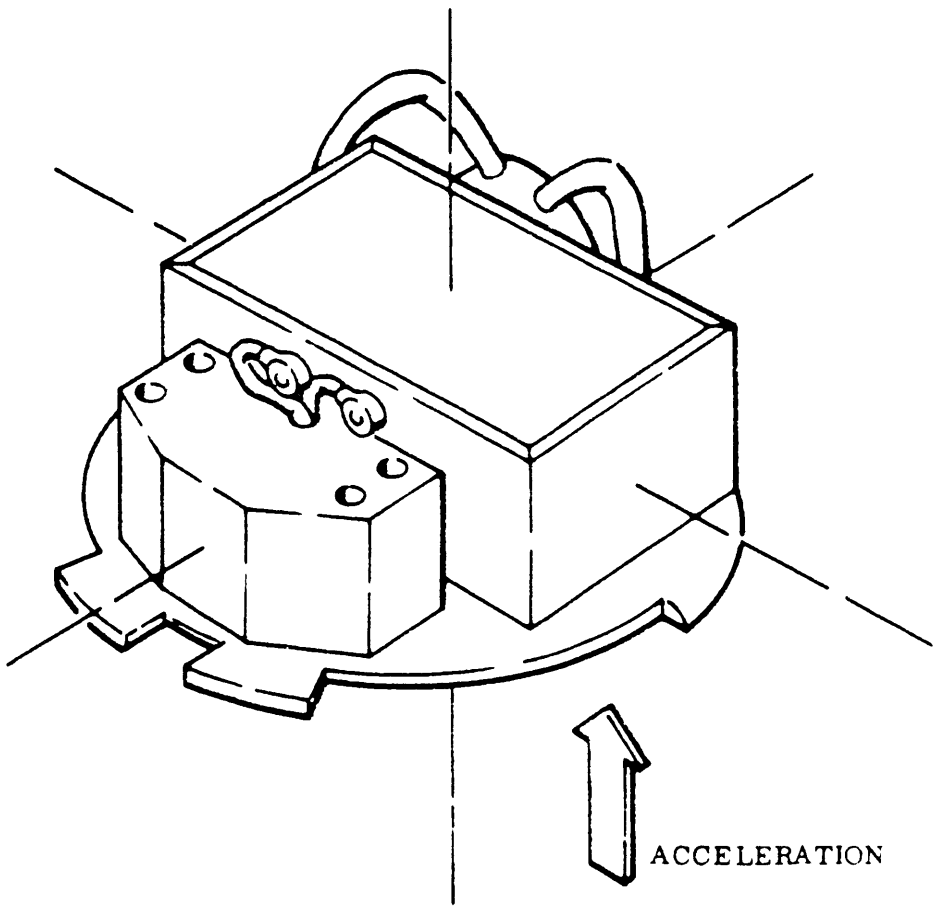


Figure 1. Environmental Testing Axes

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the programmer shall be classified as follows:

a. Preproduction and periodic production inspection (See 4.5)
b. Quality conformance inspection (See 4.6)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order, the number of product in each lot shall be three hundred (300).

4.3.2 Preproduction sample. A preproduction sample of ten (10) programmers, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the programmer by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.3 Periodic production sample. From each succeeding lot of three hundred (300) or less, a random sample of ten (10) programmers shall be selected. These sample units shall be subjected to the tests of 4.5. The programmers shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.5. (See Table III.) When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

a. Determine the cause of the defect
b. Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped
c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

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Table III. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities Preproduction	Periodic
Quality Conformance Inspection	4.6	All 10 sample units	All 10 sample units
Temperature, Storage Condition	4.5.1.1	7 of above 10	7 of above 10
Humidity	4.5.1.2	same 7 units	same 7 units
Quality Conformance Inspection	4.6	same 7 units	same 7 units
Shock, Standby Condition	4.5.1.3	same 7 units	same 7 units
Vibration, Standby Condition	4.5.1.4	same 7 units	same 7 units
Quality Conformance Inspection	4.6	same 7 units	same 7 units
Vibration, Operating Condition	4.5.1.5	same 7 units	same 7 units
Quality Conformance Inspection	4.6		
Channel One Output	4.6.1.4.4*	same 7 units	same 7 units
Channel Two Output	4.6.1.4.5*	same 7 units	same 7 units
Temperature, Operating Condition	4.5.1.6		
Low Extreme	4.5.1.6.1	same 7 units	same 7 units
Quality Conformance Inspection	4.6		
Operation	4.6.1.4*	same 7 units	same 7 units
High Extreme	4.5.1.6.2	same 7 units	same 7 units
Quality Conformance Inspection	4.6		
Operation	4.6.1.4*	same 7 units	same 7 units
Quality Conformance Inspection	4.6*	same 7 units	same 7 units
Acceleration	4.5.1.7	same 7 units	same 7 units
Quality Conformance Inspection	4.6		
Circuit Resistance	4.6.1.1	same 7 units	same 7 units
Arming Circuit	4.6.1.3.1	same 7 units	same 7 units
Enclosure	4.6.1.5	same 7 units	same 7 units
Operating Life	4.5.1.8	1 of remaining 3	1 of remaining 3
Quality Conformance Inspection	4.6		
Operation	4.6.1.4*	same 1 as above	same 1 as above

*Paragraph 4.6.1.4.6 to be conducted only at the required temperature of the controlling test.

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Table III. Preproduction and Periodic Production Test Sequence (Continued)

Test	Paragraph Number	Test Quantities Preproduction	Periodic
)uality Conformance Inspection	4.6	same 1 as above	same 1 as above
'ungus	4.5.1.9	1 of remaining 2	---
)uality Conformance Inspection	4.6	same 1 as above	---
alt Spray	4.5.1.10	last 1 of 10	remaining 2 units
uality Conformance Inspection	4.6	same 1 as above	same 2 as above

4.3.4 Quality conformance inspection sampling. Each individual programmer shall be inspected as specified in 4.6.1 to determine its acceptability.

4.4 Test equipment and test conditions.

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests:

Equipment	Capability
a. Resistance measuring device	10 to 20,000 ohms; reading accuracy of plus or minus 5 percent; applied current limited to a maximum of 10 milliamperes
b. Resistance measuring device	0.01 to 30.00 ohms; reading accuracy of plus or minus 5 percent; applied current limited to a maximum of 10 milliamperes
c. Megohmmeter	5 to 20 megohms; accuracy 5 percent with 500 plus or minus 50 volts DC potential

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Equipment	Capability
d. Temperature chamber	Maintain temperature at: Plus 35 plus 5 minus 0 degrees F Plus 120 plus 0 minus 5 degrees F
e. Oscilloscope	5-inch diameter tube, triggered, calibrated sweep of 1 millisecond per centimeter, rise time less than 1 microsecond; camera attached for recording trace
f. Tester, Acceptance, Range and Airframe Separation Programmer	BuWeps LD 541505. Operate in accordance with OD 12995.
g. Mass spectrometer helium leak detector	Minimum sensitivity of 10 to the minus sixth power cubic centimeters per second of helium

4.4.2 Preproduction and periodic production test equipment. The following items of test equipment, in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs, are required to perform the preproduction and periodic production tests:

Equipment	Capability
a. Temperature chamber	Maintain temperature at: Minus 65 plus 5 minus 0 degrees F Plus 35 plus 5 minus 0 degrees F Plus 120 plus 0 minus 5 degrees F Plus 160 plus 0 minus 5 degrees F
b. Shock equipment	Apply shocks of 40 plus or minus 4g peak magnitude; a one-half sine wave pulse shape of 5 plus 3 minus 0 milliseconds time duration; maintain direction along specified axes
c. Vibration testing machine	Control frequency from 20 to 2000 cps while subjecting the test item to 0.10-inch double amplitude or 10g acceleration, whichever is less; maintain direction of vibration along specified axes
d. Centrifuge	0 to 27.5g; maintain acceleration at 25 plus 2.5 minus 0g

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4.4.3 Test conditions. Unless otherwise specified, the programmer shall be subjected to the preproduction and periodic production tests and the quality conformance inspection tests under the following conditions:

a. Temperature	Room ambient (65 to 95 degrees F)
b. Altitude	Normal ground atmospheric pressure
c. Humidity (Relative)	Room ambient up to 95 percent

4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples shall be subjected to the tests specified in 4.5.1 after satisfactorily passing the quality conformance inspection detailed in 4.6.

4.5.1 Test procedure. Subject the programmer to the test sequence listed in Table III. Unless otherwise specified in the individual test paragraph, the quality conformance inspection as specified in Table III shall be performed subsequent to environmental tests under test conditions specified in 4.4.3. Failure of the programmer to pass any one of these tests shall constitute a defective unit.

4.5.1.1 Temperature, storage condition. The programmer shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours and maintained at this temperature for not less than 48 hours. The unit shall then be subjected to room temperature for not less than 4 hours prior to performing subsequent tests.

4.5.1.2 Humidity. The programmer shall be subjected to Humidity Test, Procedure I as specified in MIL-E-5272. Connector 1A1J2 shall be sealed against moisture during this test.

4.5.1.3 Shock, standby condition. Subject the programmer to shocks of 40 plus or minus 4g peak magnitude with a pulse shape approximating a one-half sine wave, and a duration of 5 plus 3 minus 0 milliseconds. Shocks shall be applied three times in each direction along each of three mutually perpendicular axes shown in Figure 1. Mount the programmer in a manner similar to that shown in Figure 2.

4.5.1.4 Vibration, standby condition. The programmer, rigidly mounted as shown in Figure 2, shall be subjected to the vibration tests for Type I equipment as specified in MIL-STD-167 up to 25 cps in each of the three axes shown in Figure 1.

4.5.1.5 Vibration, operating condition. The programmer, rigidly mounted as shown in Figure 2, shall be subjected to vibration from 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less, in each of the three axes shown in Figure 1. The frequency range shall be traversed in not less than 2.5 minutes, once in each axis. Perform tests a, b, and c as specified in 4.6.1.4.4 and

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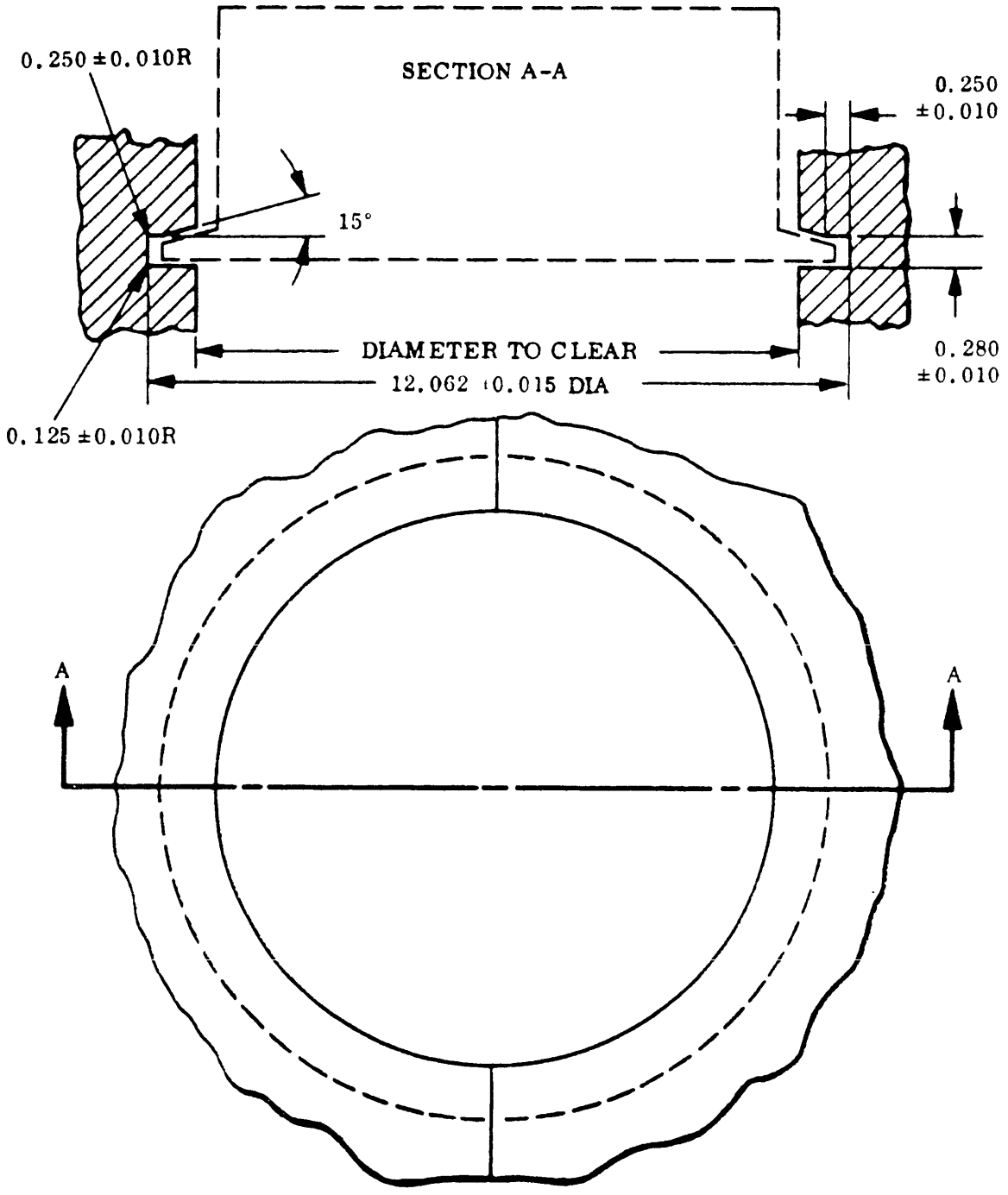


Figure 2. Progammer Mounting for Shock, Vibration and Acceleration

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4.6.1.4.5, only at ambient temperature, not less than twice in each axis. The readback time tolerance of each individual reading shall be plus or minus 0.05 second for channel one and plus or minus 0.80 second for channel two.

4,5.1.6 Temperature, operating condition.

4.5.1.6.1 Low extreme. The programmer shall be subjected to a temperature of plus 35 plus 5 minus 0 degrees F for not less than 4 hours, and while at that temperature, the programmer shall pass the operation tests of 4.6.1.4.

4.5.1.6.2 High extreme. The programmer shall be subjected to a ternperature of 120 plus 0 minus 5 degrees F for not less than 4 hours and while at that temperature shall pass the operation tests of 4.6.1.4.

4.5.1.7 Acceleration. The programmer, mounted as shown in Figure 2, shall be subjected to acceleration of 25 plus 2.5 minus 0g in the direction shown in Figure 1 for not less than 4 seconds. Thereafter, the programmer shall pass the quality conformance inspection of 4.6.1.1, 4.6.1.3.1, and 4.6.1.5.

4.5.1.8 Operating life. The programmer shall be energized as described in 4. 6.1.4 for a period of 500 hours at room ambient temperature. The operation of the programmer shall be tested as specified in 4.6.1.4.1 through 4.6. 1.4.8 at least once during every 50-hour period; these tests shall be performed at room ambient temperature only. At the end of the 500-hour period the programmer shall pass the quality conformance inspection.

4.5.1.9 Fungus. Unless otherwise specified in the contract or order, this test shall be performed on preproduction sample units only. The programmer shall be subjected to Fungus Resistance Tests, Procedure I, as specified in MIL-E-5272 .

4.5.1010 Salt spray. The programmer shall be subjected to Salt Spray Tests, Procedure I, as specified in MIL-E-5272 for 50 hours. Connector 1A1J2 shall be sealed against moisture during this test.

4.6 Quality conformance inspection. Each programmer shall be subjected to the tests and examinations as described in 4.6.1.

4.6.1 Test procedure. The programmer shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the programmer to pass the tests and examinations shall constitute a failure.

4.6.1.1 Circuit resistance. Using the two resistance measuring devices specified in 4.4.la and b, measure the resistance of the programmer circuits listed in Table I. The resistance shall be an indicated therein.

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4.6.1.2 Insulation resistance. Using the megohmmeter specified in 4.4.1c, measure the resistance between:

a. Each of the eight independent circuits of connector 1A1J2
b. The eight independent circuits of connector 1A1J2 and the programmer housing
c. The eight independent circuits of connector 1A1J2 and the shell of connector 1A1J2

When a potential of 500 plus or minus 50 volts DC is applied, the resistances shall be not less than 20 megohms.

4.6.1.3 Nonfunction indication.

4.6.1.3.1 Arming circuit. Using the tester specified in 4.4.1f, measure the resistance between contacts S and D of connector 1A1J1. After 30 seconds, the resistance shall be not less than 80 ohms. After the acceleration test of 4.5.1.7, the resistance shall be not more than 80 ohms.

4.6.1.3.2 Interlock circuit. Using the tester specified in 4.4.1f, measure the resistance between contacts S and A of connector 1A1J1. After 30 seconds, the resistance shall be not less than 80 ohms.

4.6.1.3.3 Misfire indicator circuit. Using the megohmmeter specified in 4.4.1c, measure the resistance between contacts Y and U of connector 1A1J1. The resistance shall be not less than 5 megohms, when a potential of 500 plus or minus 50 volts DC is applied.

4.6.1.4 Operation. Using the tester specified in 4.4.1f, energize the programmer with 12.75 plus or minus 0.25 volts DC applied between contacts H (negative) and f (positive) of connector 1A1J1. The programmer shall then provide the voltages indicated in the following tests.

4.6.1.4.1 Regulated voltage. Using the tester specified in 4.4.1f, measure the voltage between contacts S and M (positive) of connector 1A1J1. The tester shall indicate 10.000 plus or minus 0.010 volts DC.

4.6.1.4.2 Resaturation voltage. Using the tester specified in 4.4.1f, measure the voltage between contacts S and L (positive) of connector 1A1J1. The tester shall indicate 0.210 plus or minus 0.010 volt DC.

4.6.1.4.3 Twenty-eight volt test. Using the tester specified in 4.4.1f, measure the voltage between contacts S and d (positive) of connector 1A1J1. The tester shall indicate 28 plus or minus 3 volts DC.

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4.6.1.4.4 Channel one output. Unless otherwise specified, this test shall be performed simultaneously with the performance tests of 4.6.1.4.6. Using the tester specified in 4.4.1f, apply in sequence across contacts S and F of connector 1A1J1, the voltages specified below. The programmer shall provide a readback voltage of 4 volts across contacts S and D of connector 1A1J1 and a readback time in accordance with Table II when channel one output occurs.

a. Apply positive saturation voltage of 10.000 plus or minus 0.010 volts DC through a 200K plus or minus 2K ohm resistor for at least 2 seconds.

b. Apply a negative channel one set-in voltage shown in Table II for 100.0 plus or minus 0.1 milliseconds duration. Rise time (10 to 90 percent) to 100-millisecond pulse shall be not more than 150 microseconds and fall time (10 to 90 percent) not more than 250 microseconds. With 50 volts set in, pulse width is measured at 25-volt level (half amplitude point). Switching time from presaturation to set-in voltage shall be not more than 1 millisecond and the circuit to the programmer shall not be open at any time during this switching operation.

c. Apply positive resaturation voltage of 2.5000 plus or minus 0.0025 volts DC until 4 volts DC is obtained between contacts of S and D of connector 1A1J1. The appearance of the 4 volts DC across. contacts S and D is indicated by stopping of the tester counter. Observe readback time on counter.

4.6.1.4.5 Channel two output. Unless otherwise specified, this test shall be performed simultaneously with the performance tests of 4.6.1.4.6. Using the tester, specified in 4.4.1f, apply in sequence across contacts S and B of connector 1A1J1, the voltages specified below. The programmer shall provide a readback voltage of 4 volts across contacts S and A of connector 1A1J1 and readback time in accordance with Table II when channel two output occurs.

a. Apply positive saturation voltage of 10.000 plus or minus 0.010 volts DC through a 500K plus or minus 5K ohm resistor for at least 2 seconds.

b. Apply a negative channel two set-in voltage noted in Table II for 100.0 plus or minus 0.1 milliseconds duration. Rise time (10 to 90 percent) to 100-millisecond pulse shall be not more than 150 microseconds and fall time (10 to 90 percent) not more than 250 microseconds. With 50 volts set in, pulse width is measured at 25-volt level (half amplitude point). Switching time from pre-saturation to set-in voltage shall be not more than 1 millisecond and the circuit to the programmer shall not be open at any time during this switching operation.

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c. Apply positive resaturation voltage of 0.210 plus or minus 0.010 volt DC until 4 volts DC is obtained at contacts S and A of connector 1A1J1. The 0.210 volt DC is supplied by the programmer across contacts S and L (positive) of connector 1A1J1. The appearance of 4 volts DC across contacts S and A is indicated by the stopping of the tester counter. Observe readback time on the counter.

4.6.1.4.6 Performance at specific temperatures. Place the programmer in the temperature chamber specified in 4.4.1d. Perform the following tests while the programmer is stabilized at each of the temperatures listed in Table II for 4 hours minimum.

a. Channel one readback time: take six consecutive readings at each channel one set-in voltage and temperature listed in Table II. Tabulate the last five of the six readings and then calculate the RMS error. (See 6.3.2.) The RMS error shall not exceed that specified in Table II.

b. Channel two readback time: take six consecutive readings at each channel two set-in voltage and temperature listed in Table II. Tabulate the last five of the six readings and then calculate the RMS error. (See 6.3.2.) The RMS error shall not exceed that specified in Table II.

4.6.1.4.7 Output energy. Energize the programmer as specified in 4.6.1.4. Connect a mercury switch and a 2 ohm, 1 watt, 1 percent resistor in series across contacts S and d of connector 1A1J1. Attach the vertical input lead of the oscilloscope specified in 4.4.1e to contact d. Set the oscilloscope vertical sensitivity control at 5 volts per centimeter. Set the oscilloscope horizontal sensitivity control at 1 millisecond per centimeter. Using the oscilloscope camera, observe and record the output energy pulse of the programmer. The output energy, as computed in 4.6.1.4.7.1, shall be not less than 2 million ergs.

4.6.1.4.7.1 Output energy calculation procedure. From the photographic record of the output energy pulse, measure the output voltage at 0.50, 1.50, and 2.50 milliseconds from the beginning of the pulse and identify the measurements as E_1, E_2, and E_3 respectively, see Figure 3. The energy content of the programmer output energy pulse is defined here as the summation of energy increments calculated from the three voltages and the three one-millisecond time intervals. The basic energy equation is:

$$\text{Energy (ergs)} = \sum \frac{E^2}{R_L} \times T \times 10^7$$

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Where: E = voltages E_1, E_2, E_3 (volts)

R_L = load resistance (ohms)

T = time interval (seconds)

10^7 = conversion factor from watt-seconds, or joules, to ergs

In this test, R_L = 2.00 ohms, and T = 0.001 second. The energy content of the pulse shall then be calculated from the following:

Energy (ergs) =

A typical programmer output energy pulse is shown in Figure 3.

4.6.1.4.8 Channel one output guarantee. Repeat channel one output test, 4.6.1.4.4, at a set-in voltage of 50.000 plus or minus 0.050 volts DC. When the voltage from contact N of connector 1A1J1 replaces the 2.5000 volts DC, specified in 4. 6.1.4.4 step c, channel one output shall occur at 8 plus or minus 1.5 seconds.

4.6.1.5 Enclosure. Using the leak detector specified in 4.4.1g, test the programmer at one atmosphere pressure differential. The leakage rate shall be not more than 10 to the minus fifth power cubic centimeter per second of helium.

4.6.1.6 Visual examination. Visual examination shall be made on each programmer submitted for delivery to the procuring activity. The programmer shall be examined for conformance to the requirements of good workmanship in 3.5. Nonconformance is cause for rejection of individual unit.

4.7 Preservation, packaging, packing, and marking. The programmer shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

5 . PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. (See 6.2b and 6.4.)

5.1.1 Level A. Cleaning, drying, preservation and packaging methods for Level A shipment shall be in accordance with the instructions contained in the applicable documents listed in BuWeps LD 541554.

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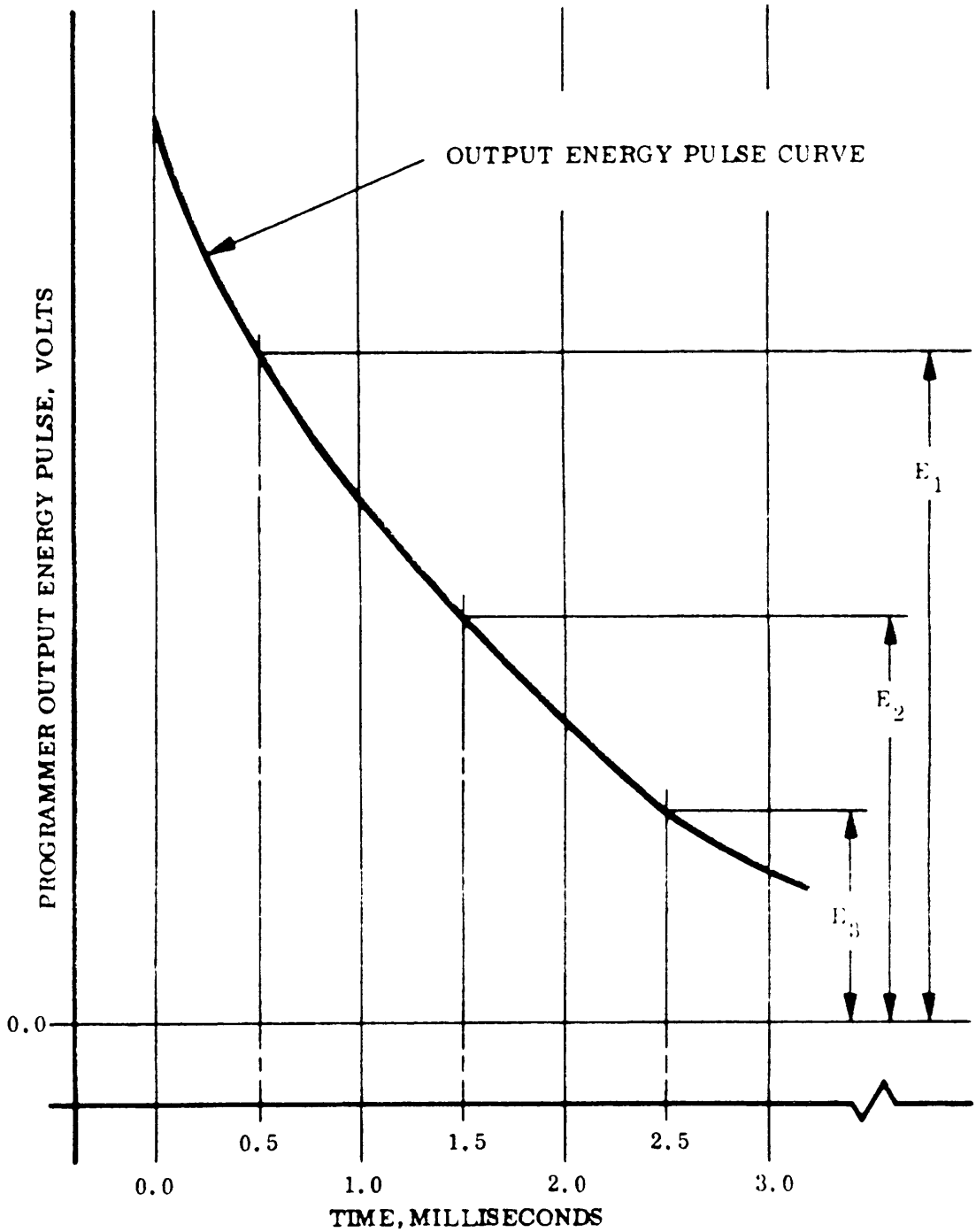


Figure 3. Typical Record of Programmer Output Energy Pulse

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5.1.2 Level C. Cleaning, drying, preservation and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C as specified. (See 6.2c and 6.4.)

5.2.1 Level A. The programmer, preserved and packaged an specified in 5.1.1, shall be packed in Container Mark 176 Mod 1, in accordance with BuWeps LD 497947 and BuWeps LD 541555.

5.2.2 Level C. The programmer, packaged as specified in 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6 . NOTES

6.1 Intended use and description.

6.1.1 Intended use. The programmer is intended to be used in an ASROC missile, and provides sequencing of separation requirements.

6.1.2 Description. The programmer is an electronic device designed to provide two signals in sequence and under conditions that have been predetermined by the Fire Control System. The first signal is produced when the programmer senses a certain velocity, and the second signal is produced after a certain time interval has elapsed. The programmer provides eight pass-through circuits allowing interconnection between the Fire Control System and the payload.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Required level of preservation and packaging
c. Required level of packing
d. Any exceptions to this specification
e. Quality conformance inspection data to be shipped with each RASP.

6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

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6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Unit condition.

6.3.1.1 Operating. The condition of the programmer when performing its required function.

6.3.1.2 Standby. Standby is the condition of the programmer when aboard ship and is:

a. Stored, handled, or transported
b. Assembled with the end item and installed on the launching mechanism prior to operation

6.3.2 RMS error. The root-mean-square (RMS) error for a sample of observation is a statistical indication of the deviation of individual observations from the theoretical correct value. It is defined for use herein as:

$$\sqrt{\frac{\sum (X - a)^2}{N}}$$

Where: X = Individual reading of readback time

a = Required readback time as given in Table II

N = Number of individual readings of readback time used in the calculation (specified in 4.6.1.4.6)

6 . 4 Preparation on for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-P-82534(OS) PROGRAMMER, RANGE AND AIRFRAME SEPARATION MK 3 MOD 1

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1426
1 APR 83

C-8979

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Naval Ordnance Systems Command
Washington, D.C. 20360

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MIL-P-82534(OS)
Amendment -2
2 May 1969
Superseding
Amendment-1
26 November 1968

MILITARY SPECIFICATION

PROGRAMMER, RANGE AND AIRFRAME SEPARATION

MARK 3 MOD 1

This amendment forms a part of Military Specification MIL-P-82534(OS) of 18 April 1968 and has been approved by the Naval Ordnance Systems Command, Department of the Navy

Wherever it appears in specification delete "Bureau of Naval Weapons" and substitute "Naval Ordnance Systems Command"; also delete "BuWeps" and substitute "NAVORD".

Page 10, Table III. Make the following changes:

From:	Humidity	4.5.1.2
	Quality Conformance Inspection	4.6
To:	Humidity	4.5.1.2
	Quality Conformance Inspection	4.6**
From:	Vibration, Standby Condition	4.5.1.4
	Quality Conformance Inspection	4.6
To:	Vibration, Standby Condition	4.5.1.4
	Quality Conformance Inspection	4.6**
From:	High Extreme	4.5.1.6.2
	Quality Conformance Inspection	4.6
	Operation	4.6.1.4
	Quality Conformance Inspection	4.6*
To:	High Extreme	4.5.1.6.2
	Quality Conformance Inspection	4.6
	Operation	4.6.1.4
	Quality Conformance Inspection	4.6* **

Add the following note below table:

**All tests of paragraph 4.6 shall be conducted except the enclosure test (4.6.1.5).

FSC 1340

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Page 12, paragraph 4.4.2(c), under "Capability". Delete and substitute "Automatically control displacement/acceleration amplitude of fundamental vibration input to RASP at 0.10-inch double amplitude or 10g, whichever is less, over the frequency range of 20 to 2000 cps. Monitor cross-axis motion of RASP in two directions."

Pages 13 and 1.5, paragraph 4.5.1.5. Delete and substitute:

"4.5.1.5 Vibration, operating condition. The programmer shall be subjected to sinusoidal vibration over the frequency range of 20 plus or minus 3, to 2000 plus or minus 40 cps, and at a double displacement amplitude of 0.100 plus or minus 0.015 inch or a zero-to-peak acceleration of 10 plus or minus 1.5g, whichever is less, in each of the three axes shown in Figure 1. The frequency range shall be traversed at a logarithmic rate in not less than 2.5 minutes, once in each axis. The vibration system used shall meet the performance requirements specified in 4.5.1.5.1 through 4.5.1.5.3. Perform tests a, b, and c, as specified in 4.6.1.4.4 and 4.6.1.4.5 at room ambient temperature only, not less than twice in each axis. Readings shall be taken at approximately equal time intervals over the entire frequency range. The readback time tolerance of each individual reading shall be plus or minus 0.05 second for channel one and plus or minus 0.80 second for channel two.

4.5.1.5.1 Programmer support requirements. For tests specified in 4.5.1.5, the programmer shall be supported only in the area shown in Figure 2 except that the free ends of the programmer payload cable and test cable may be supported.

4.5.1.5.2 Vibration system requirements. The vibration system used to conduct tests of 4.5.1.5 shall have a continuous duty force rating of at least 125 percent of the calculated force rating required to vibrate the programmer and supporting fixtures at specified levels. Total harmonic distortion in the acceleration waveform over the test range shall not exceed 10 percent. The acceleration system noise level at the unloaded vibration machine exciter shall be less than 0.4g rms with exciter power supply and field energized, and oscillator gain control set for maximum attenuation. The displacement system noise level under the foregoing conditions shall not exceed 0.005-inch double amplitude.

4.5.1.5.3 Vibration control and monitor requirements

4.5.1.5.3.1 Vibration machine output. Output of vibration machine shall be controlled by an automatic amplitude servo system and shall be maintained at the levels of displacement or acceleration specified in 4.5.1.5 by controlling the amplitude of the fundamental frequency.

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Any distortion of the fundamental signal, regardless of source, shall not be allowed to affect the measurement of displacement or acceleration amplitude of the fundamental input frequency to the vibration machine. Such distortion shall be eliminated from the input control signal by use of a tracking band pass filter between the input control circuit and the amplitude servo system.

4.5.1.5.3.2 Vibration pickups. During tests specified in 4.5.1.5 the output of the vibration machine shall be controlled and monitored by three piezoelectric accelerometers mounted in the vicinity of the programmer so as to indicate the greatest motion of the supporting fixture on which the programmer is directly mounted. The accelerometers shall be positioned in three mutually orthogonal axes. One accelerometer, mounted with its axis parallel to the axis of motion, shall be used to control power to the vibration machine. The remaining two accelerometers, mounted parallel and perpendicular to the vibration machine trunnion axis, shall be used to monitor cross-axis motion. Unfiltered cross-axis motion as indicated by either accelerometer shall not exceed the filtered input to the vibration machine. Motion transducers other than piezoelectric accelerometers shall not be used for input control or monitoring.

4.5.1.5.3.3 Control and monitor circuits. Circuits used to control and monitor the vibration system for tests specified in 4.5.1.5 shall meet the following requirements:

(a) Displacement circuits shall indicate amplitude of input displacement to the programmer over the range of 0.010-inch double amplitude to the upper limit of the equipment to an accuracy of plus or minus 10 percent from 10 to 100 cps.

(b) Acceleration circuits shall indicate amplitude of input acceleration to the programmer to an accuracy of plus or minus 10 percent over the range of 17 to 2040 cps. The circuits shall be capable of indicating the amplitude of both the fundamental and. unfiltered signal. It shall not be necessary, however, to indicate these signals simultaneously."

Page 15, paragraph 4.6.1.1, last sentence: Delete "an indicated" and substitute "as indicated".

Project No. 1340-N268

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NOTICE OF INACTIVATION
FOR NEW DESIGN

MIL-P-82534(OS)
NOTICE 1
15 SEPTEMBER 1986

MILITARY SPECIFICATION

PROGRAMMER, RANGE AND AIRFRAME SEPARATION MARK 3 MOD 1

This notice should be filed in front of MIL-P-82534(OS) dated 18 April 1968 with Amendment 2 dated 2 May 1969.

MIL-P-82534(OS) is inactive for new design and is no longer used by the Navy except for replacement purposes.

Preparing Activity:
Navy - 0S
Project No. 1340-N717

AMSC N/A FSC 1340

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